

Samundar Singh

Software Engineer

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PROFESSIONAL EXPERIENCE

Caterpillar Inc, (Associate Software Engineer)

Aug 2022 – present | Chennai

- Created reusable Angular components and directives, streamlining development processes across multiple applications.ensuring 80-90% component reusability across project.
- Collaborated with cross-functional teams to analyze project requirements and successfully implementing user-centric features across 33 modules of Caterpillar's business application.
- Utilized **RxJS** for reactive programming and managing asynchronous data. Assisted in designing dynamic request forms with edit and view capabilities based on payload structures.
- Developed a chatbot using **Ollama** model to convert natural language to SQL queries, **engineered prompts** and custom datasets based on schema, and **fine-tuned** the model to achieve over 86% accuracy.
- Enhanced the Data Management System by addressing security vulnerabilities raised by **bug-bounty**, fixing low- and mid-level security bugs, and implementing **HTTP encryption** using the crypto-js npm package in an **HTTP interceptor** to protect data in URLs and payloads, resulting in a 98% secure application and a 90% improvement in system security.

Caterpillar, (Internship)

Jan 2022 – Jul 2022

- Engineered real-time dashboards for mining machinery using Angular's **NgRx** state management, improving data handling efficiency by 40% and reducing dashboard load time by 30%, ensuring real-time updates.
- **Utilized web sockets, and Plotly** to build 15+ dashboards with interactive data visualizations, improving real-time insights, and Integrated support for generating and sharing machine statuses via various graphs.
- Implemented a **report creator** module with **drag-and-drop** functionality, enabling seamless machine report generation through an intuitive user interface.

SKILLS

Angular | Python | JavaScript | PostgreSQL | MongoDB | REST API | Node.js | Typescript | Django | Data Structure and Algorithm | Problem Solving | Object Oriented Programming

EDUCATION

Master of Computer Application | (CGPA - 9.3),

2019 – 2022

College of Engineering Guindy, Anna University

Bachelor of Science (Computer Application) | (CGPA - 7.7),

2015 – 2018

Ranchi College, Ranchi University

PROJECTS

Smart Agriculture Dashboard with Predictive Analytics | Python, Angular ✍

- A web-based application **1M+ data points** to forecast temperature variations, enhancing decision-making capabilities for farm management.
- Attained **93% predictive accuracy** by applying **linear and multiple regression models**, coupled with rigorous outlier removal.
- Provided users with actionable insights through dynamic and visually appealing **graph-based representations** of prediction data, improving data interpretation speed by 60% and decision-making efficiency by 45%.

Result Management System | Java, postgres, Java-fx ✍

- Developed a marks distribution and attendance management application with 4 modules using **javafx**.
- Handled 20+ data transactions using **ORM** in a **relational database**.
- Supported **CRUD** operations and integrated **PDF generation** for results, improving data accessibility by 50% and enhancing record-keeping efficiency by 40%.
- Secured **85% test coverage**, guaranteeing high functionality and dependable performance.